

Radiography Physics Quick Reference

Independent study resource - verify course-specific conventions and current institutional guidance.

Exposure relationships

Concept	Key point
mAs	$\text{mA} \times \text{time} = \text{mAs}$. Photon quantity changes approximately in direct proportion to mAs.
kVp	Primarily affects beam energy and penetration; also affects photon quantity and receptor exposure.
15% rule	A roughly 15% increase in kVp can approximately double receptor exposure under traditional technique relationships; reducing mAs by half can help maintain receptor exposure.

Geometry

Concept	Key point
Inverse square law	Intensity is inversely proportional to the square of distance: $I_1 / I_2 = D_2^2 / D_1^2$.
Direct square law	When SID changes and receptor exposure must be maintained: $\text{mAs}_2 / \text{mAs}_1 = \text{SID}_2^2 / \text{SID}_1^2$.
OID	Increasing object-to-image receptor distance generally increases magnification and decreases recorded spatial detail.
SID	Increasing source-to-image distance generally reduces magnification and beam intensity at the receptor.

Image quality

Concept	Key point
Spatial resolution	Influenced by focal spot size, motion, OID, SID, detector sampling, and system processing.
Contrast	Subject contrast depends heavily on attenuation differences and kVp; displayed contrast can be altered digitally.
Scatter	Compton interactions are the major source of scatter in diagnostic imaging. Collimation reduces scatter and patient dose.

Radiation protection

Concept	Key point
ALARA	As Low As Reasonably Achievable.
Time	Reduce time near a radiation source when possible.
Distance	Increase distance; scatter intensity falls rapidly with distance.
Shielding / barriers	Use appropriate structural or personal protection according to current policy and procedure.